

Creation Date 11-Feb-2010

Revision Date 02-Feb-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                              |
|---------------------------|----------------------------------------------|
| Product Description:      | <b>Furfuryl alcohol</b>                      |
| Cat No. :                 | <b>A10968</b>                                |
| Synonyms                  | 2-Furanmethanol                              |
| Index No                  | 603-018-00-2                                 |
| CAS No                    | 98-00-0                                      |
| EC No                     | 202-626-1                                    |
| Molecular Formula         | C <sub>5</sub> H <sub>6</sub> O <sub>2</sub> |
| REACH registration number | -                                            |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

[begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

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## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Acute oral toxicity                                  | Category 3 (H301) |
| Acute dermal toxicity                                | Category 3 (H311) |
| Acute Inhalation Toxicity - Vapors                   | Category 2 (H330) |
| Skin Corrosion/Irritation                            | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319) |
| Carcinogenicity                                      | Category 2 (H351) |
| Specific target organ toxicity - (single exposure)   | Category 3 (H335) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373) |

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H319 - Causes serious eye irritation  
H315 - Causes skin irritation  
H335 - May cause respiratory irritation  
H330 - Fatal if inhaled  
H373 - May cause damage to organs through prolonged or repeated exposure if inhaled  
H351 - Suspected of causing cancer if inhaled  
H301 + H311 - Toxic if swallowed or in contact with skin  
Combustible liquid

#### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor/physician  
P362 - Take off contaminated clothing and wash before reuse  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

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## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component        | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                          |
|------------------|---------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol | 98-00-0 | EEC No. 202-626-1 | >95      | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Eye Irrit. 2 (H319)<br>Skin Irrit. 2 (H315)<br>Carc. 2 (H351)<br>STOT SE 3 (H335)<br>STOT RE 2 (H373) |

REACH registration number

-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

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## **5.1. Extinguishing media**

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## **5.2. Special hazards arising from the substance or mixture**

Very toxic by inhalation. Combustible material. Risk of ignition. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

## **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Evacuate personnel to safe areas. Take precautionary measures against static discharges.

### **6.2. Environmental precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Keep under nitrogen.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1A

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1

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<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component        | European Union | The United Kingdom | France                                                                               | Belgium                                                                                                                            | Spain                                                                                                                                                                             |
|------------------|----------------|--------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol |                |                    | TWA / VME: 10 ppm (8 heures).<br>TWA / VME: 40 mg/m <sup>3</sup> (8 heures).<br>Peau | TWA: 10 ppm 8 uren<br>TWA: 41 mg/m <sup>3</sup> 8 uren<br>STEL: 15 ppm 15 minuten<br>STEL: 61 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 15 ppm (15 minutos).<br>STEL / VLA-EC: 61 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 5 ppm (8 horas)<br>TWA / VLA-ED: 20 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component        | Italy | Germany | Portugal                                               | The Netherlands | Finland                                                                                                                                           |
|------------------|-------|---------|--------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol |       | Haut    | STEL: 15 ppm 15 minutos<br>TWA: 10 ppm 8 horas<br>Pele |                 | TWA: 2 ppm 8 tunteina<br>TWA: 8.1 mg/m <sup>3</sup> 8 tunteina<br>STEL: 10 ppm 15 minuutteina<br>STEL: 41 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component        | Austria                                                                     | Denmark                                                                                                                              | Switzerland                                                                                                                                   | Poland                                                                          | Norway                                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol | Haut<br>MAK-TMW: 5 ppm 8 Stunden<br>MAK-TMW: 20 mg/m <sup>3</sup> 8 Stunden | TWA: 5 ppm 8 timer<br>TWA: 20 mg/m <sup>3</sup> 8 timer<br>STEL: 10 ppm 15 minutter<br>STEL: 40 mg/m <sup>3</sup> 15 minutter<br>Hud | Haut/Peau<br>STEL: 10 ppm 15 Minuten<br>STEL: 40 mg/m <sup>3</sup> 15 Minuten<br>TWA: 10 ppm 8 Stunden<br>TWA: 40 mg/m <sup>3</sup> 8 Stunden | STEL: 60 mg/m <sup>3</sup> 15 minutach<br>TWA: 30 mg/m <sup>3</sup> 8 godzinach | TWA: 5 ppm 8 timer<br>TWA: 20 mg/m <sup>3</sup> 8 timer<br>STEL: 10 ppm 15 minutter. value calculated<br>STEL: 30 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |

| Component        | Bulgaria | Croatia                                                              | Ireland                                                                                                                  | Cyprus | Czech Republic                                                                                               |
|------------------|----------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol |          | TWA-GVI: 10 ppm 8 satima.<br>TWA-GVI: 40 mg/m <sup>3</sup> 8 satima. | TWA: 10 ppm 8 hr.<br>TWA: 40 mg/m <sup>3</sup> 8 hr.<br>STEL: 15 ppm 15 min<br>STEL: 60 mg/m <sup>3</sup> 15 min<br>Skin |        | TWA: 20 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 40 mg/m <sup>3</sup> |

| Component        | Estonia                                                                                                                                           | Gibraltar | Greece                                                                                                                              | Hungary                                                                                                                          | Iceland                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol | Nahk<br>TWA: 5 ppm 8 tundides.<br>TWA: 20 mg/m <sup>3</sup> 8 tundides.<br>STEL: 10 ppm 15 minutites.<br>STEL: 40 mg/m <sup>3</sup> 15 minutites. |           | skin - potential for cutaneous absorption<br>STEL: 15 ppm<br>STEL: 60 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 40 mg/m <sup>3</sup> | STEL: 40 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 20 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges bőrön keresztüli felszívódás | TWA: 5 ppm 8 klukkustundum.<br>TWA: 20 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 10 ppm<br>Ceiling: 40 mg/m <sup>3</sup> |

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| Component        | Latvia                     | Lithuania                                                                                              | Luxembourg | Malta | Romania                                                                                                                         |
|------------------|----------------------------|--------------------------------------------------------------------------------------------------------|------------|-------|---------------------------------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol | TWA: 0.5 mg/m <sup>3</sup> | TWA: 5 ppm IPRD<br>TWA: 20 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 10 ppm<br>STEL: 40 mg/m <sup>3</sup> |            |       | TWA: 12.5 ppm 8 ore<br>TWA: 50 mg/m <sup>3</sup> 8 ore<br>STEL: 25 ppm 15<br>minute<br>STEL: 100 mg/m <sup>3</sup> 15<br>minute |

| Component        | Russia                                      | Slovak Republic                                                                   | Slovenia | Sweden                                                                                                                                                                           | Turkey |
|------------------|---------------------------------------------|-----------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Furfuryl alcohol | Skin notation<br>MAC: 0.5 mg/m <sup>3</sup> | Potential for cutaneous<br>absorption<br>TWA: 10 ppm<br>TWA: 41 mg/m <sup>3</sup> |          | Indicative STEL: 10 ppm<br>15 minuter<br>Indicative STEL: 40<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 5 ppm 8 timmar.<br>NGV<br>TLV: 20 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

| Component                           | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|-------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Furfuryl alcohol<br>98-00-0 ( >95 ) |                                 |                                    |                                   | DNEL = 4mg/kg bw/day                 |

| Component                           | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|-------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Furfuryl alcohol<br>98-00-0 ( >95 ) | DNEL = 8mg/m <sup>3</sup>           | DNEL = 143mg/m <sup>3</sup>            | DNEL = 8mg/m <sup>3</sup>             | DNEL = 31mg/m <sup>3</sup>               |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                           | Fresh water     | Fresh water<br>sediment             | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)               |
|-------------------------------------|-----------------|-------------------------------------|--------------------|---------------------------------------|----------------------------------|
| Furfuryl alcohol<br>98-00-0 ( >95 ) | PNEC = 0.17mg/L | PNEC =<br>0.861mg/kg<br>sediment dw | PNEC = 1.7mg/L     |                                       | PNEC =<br>0.0724mg/kg soil<br>dw |

| Component                           | Marine water     | Marine water<br>sediment             | Marine water<br>Intermittent | Food chain               | Air |
|-------------------------------------|------------------|--------------------------------------|------------------------------|--------------------------|-----|
| Furfuryl alcohol<br>98-00-0 ( >95 ) | PNEC = 0.017mg/L | PNEC =<br>0.0861mg/kg<br>sediment dw |                              | PNEC = 35.3mg/kg<br>food |     |

## 8.2. Exposure controls

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## Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Goggles (European standard - EN 166)

### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                           |                                                 |                                          |
|---------------------------|-------------------------------------------------|------------------------------------------|
| Physical State            | Liquid                                          |                                          |
| Appearance                | Yellow                                          |                                          |
| Odor                      | No information available                        |                                          |
| Odor Threshold            | No data available                               |                                          |
| Melting Point/Range       | -29 °C / -20.2 °F                               |                                          |
| Softening Point           | No data available                               |                                          |
| Boiling Point/Range       | 170 °C / 338 °F                                 | @ 760 mmHg                               |
| Flammability (liquid)     | Combustible liquid                              | On basis of test data                    |
| Flammability (solid,gas)  | Not applicable                                  | Liquid                                   |
| Explosion Limits          | <b>Lower</b> 1.8 vol%<br><b>Upper</b> 16.3 vol% |                                          |
| Flash Point               | 65 °C / 149 °F                                  | <b>Method -</b> No information available |
| Autoignition Temperature  | 391 °C / 735.8 °F                               |                                          |
| Decomposition Temperature | No data available                               |                                          |
| pH                        | No information available                        | 30% aq.sol                               |
| Viscosity                 | 5 cP @ 25°C                                     |                                          |

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|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Water Solubility                        | No information available |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Component                               | log Pow                  |             |
| Furfuryl alcohol                        | 0.28                     |             |
| Vapor Pressure                          | 0.53 mbar @ 20°C         |             |
| Density / Specific Gravity              | 1.13                     |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Particle characteristics                | Not applicable (liquid)  |             |

## 9.2. Other information

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | C5 H6 O2                               |
| Molecular Weight     | 98.10                                  |
| Explosive Properties | explosive air/vapour mixtures possible |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions. Air sensitive.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.  
Excess heat. Exposure to air.

### 10.5. Incompatible materials

Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 2 |

| Component        | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation                                              |
|------------------|--------------------------|-----------------------------|--------------------------------------------------------------|
| Furfuryl alcohol | LD50 = 110 mg/kg ( Rat ) | LD50 = 657 mg/kg ( Rabbit ) | LC50 = 1.18 mg/L ( Rat ) 4 h<br>LC50 = 1.48 mg/L ( Rat ) 4 h |

|                                |            |
|--------------------------------|------------|
| (b) skin corrosion/irritation; | Category 2 |
|--------------------------------|------------|



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(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component        | EU | UK | Germany | IARC     |
|------------------|----|----|---------|----------|
| Furfuryl alcohol |    |    |         | Group 2B |

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

(i) STOT-repeated exposure;

Category 2

Target Organs

Nasal Cavities.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Harmful to aquatic organisms.

| Component        | Freshwater Fish                                      | Water Flea          | Freshwater Algae |
|------------------|------------------------------------------------------|---------------------|------------------|
| Furfuryl alcohol | LC50: = 32 mg/L, 96h static<br>(Pimephales promelas) | EC50 = 328 mg/L 24h |                  |

### 12.2. Persistence and degradability

Persistence  
Degradation in sewage  
treatment plant

No information available  
Persistence is unlikely.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| Furfuryl alcohol | 0.28    | No data available             |

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|                                                                                                               |                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>12.4. Mobility in soil</b>                                                                                 | No information available .                                                                                                       |
| <b>12.5. Results of PBT and vPvB assessment</b>                                                               | Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).       |
| <b>12.6. Endocrine disrupting properties</b><br><b>Endocrine Disruptor Information</b>                        | This product does not contain any known or suspected endocrine disruptors                                                        |
| <b>12.7. Other adverse effects</b><br><b>Persistent Organic Pollutant</b><br><b>Ozone Depletion Potential</b> | This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>13.1. Waste treatment methods</b>       |                                                                                                                                                                                                                                                                                                            |
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                                                                                                                                       |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN2874           |
| <b>14.2. UN proper shipping name</b>    | Furfuryl alcohol |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | III              |

### ADR

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN2874           |
| <b>14.2. UN proper shipping name</b>    | Furfuryl alcohol |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | III              |

### IATA

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN2874           |
| <b>14.2. UN proper shipping name</b>    | Furfuryl alcohol |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | III              |

|                                    |                       |
|------------------------------------|-----------------------|
| <b>14.5. Environmental hazards</b> | No hazards identified |
|------------------------------------|-----------------------|

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**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component        | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Furfuryl alcohol | 98-00-0 | 202-626-1 | -      | -   | X     | X    | KE-17364 | X    | X    |

| Component        | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Furfuryl alcohol | 98-00-0 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component        | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Furfuryl alcohol | 98-00-0 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Seveso III Directive (2012/18/EC)

| Component        | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Furfuryl alcohol | 98-00-0 | Not applicable                                                                            | Not applicable                                                                           |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**

Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

#### National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

#### WGK Classification

See table for values

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| Component        | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|------------------|---------------------------------------|-------------------------|
| Furfuryl alcohol | WGK1                                  |                         |

| Component        | France - INRS (Tables of occupational diseases)             |
|------------------|-------------------------------------------------------------|
| Furfuryl alcohol | Tableaux des maladies professionnelles (TMP) - RG 74, RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H330 - Fatal if inhaled

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H351 - Suspected of causing cancer

H335 - May cause respiratory irritation

H373 - May cause damage to organs through prolonged or repeated exposure

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

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|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Creation Date    | 11-Feb-2010                                        |
| Revision Date    | 02-Feb-2024                                        |
| Revision Summary | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**